CS23532-COMPUTER NETWORKS-LAB MANUAL

Practical -3

Aim

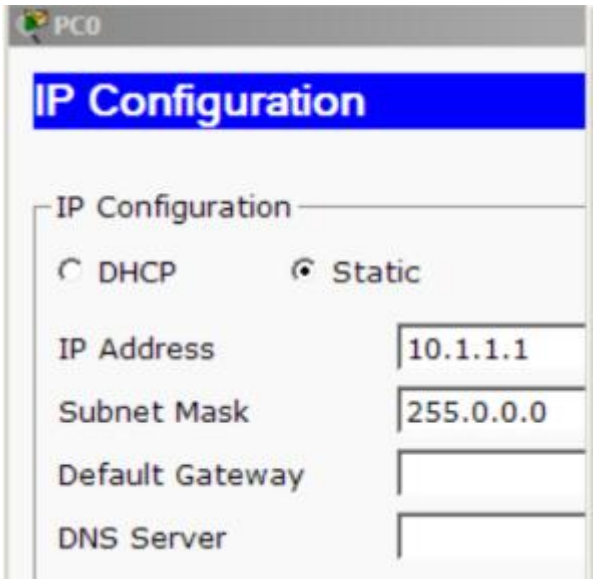
To understand the Cisco Packet Tracer environment, design simple computer networks using hubs and switches, simulate communication between devices, and identify the topology used in the college network.

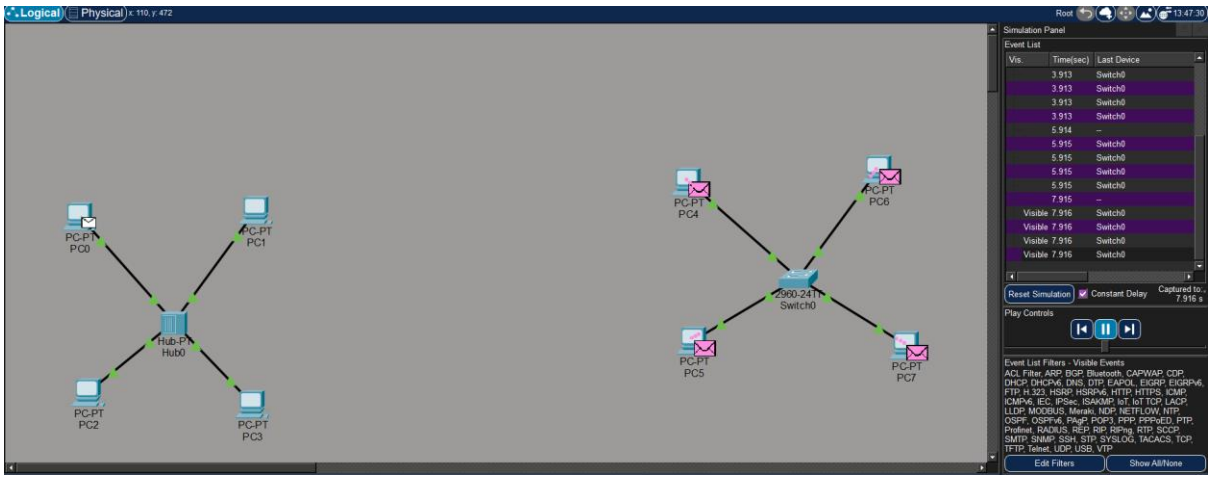
Procedure

1. Open Cisco Packet Tracer and become familiar with its workspace, device library, connections, and simulation mode.
2. **Network using Hub:**
    - Place one hub and connect 4–6 PCs to it using copper straight-through cables.
    - Assign a unique IP address to each PC in the same network.
    - Use the ping command to test communication between the PCs.
    - Observe how the hub forwards incoming data to all connected ports.
3. **Network using Switch:**
    - Create another network using one switch and 4–6 PCs.
    - Connect each PC to the switch using copper straight-through cables.
    - Assign IP addresses in the same network.
    - Test connectivity using the ping command.
    - Observe how the switch forwards frames based on the destination MAC address.
4. **Network using Both Hub and Switch:**
    - Connect a hub and a switch to form a small network containing multiple PCs.
    - Assign suitable IP addresses to all PCs.
    - Test communication between devices using ping.
    - Use Simulation Mode to observe how packets travel through the network.
5. **Comparison:**
    - Observe the behavior of the hub and switch during data transmission.

- A hub broadcasts incoming data to all its ports, whereas a switch forwards frames to the appropriate destination port when its MAC address table is available.
- Record the differences in performance and data forwarding.

6. **College Network Topology:**
   - Study the network arrangement used in the college.
   - Identify devices such as computers, switches, routers, and access points.
   - Draw and label the observed network topology in the observation book.





Result

The Cisco Packet Tracer environment was studied successfully. Networks using a hub, a switch, and both devices were designed and tested. Their data-forwarding behavior was observed and compared, and the college network topology was studied and recorded.